

Honors Metrics TA Session: Week 4

# F-Statistics, Applied Orthogonalization, and Exam Practice

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April 15, 2024

## Logistical reminders and ice-breaker

- Office hours in SHFE 201 on Wednesdays from 5:30-6:30
- TA sessions here, now
- Exam on Thursday
  - ▶ Material is up through the end of the posted notes
  - ▶ Any logistical questions?
- Go around (save til end): Name, question about the material if you have one?

# Outline for today

1. Regression inference: F-statistic derivation
2. FWL in practice
  - ▶ BLP interpretation
  - ▶ An example from economics literature
  - ▶ Sample R code
3. Some key practice problems:
  - ▶ Conditional expectation
  - ▶ Regression and independence
  - ▶ Consistency over ranges

## Review of $\chi^2$ distribution

- $\chi^2(k)$  is distribution of sum of  $k$  standard normals
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    - ★ Mean becomes  $ck$  and variance  $2c^2k$

## Important case of $\chi^2$ under normality

- Let

$$Y = X'\beta + e \quad \text{with} \quad Y|X \sim \mathcal{N}(X'\beta, \sigma^2)$$

Immediately, this implies  $e|X \sim \mathcal{N}(0, \sigma^2)$

- Then, our  $t$ -statistic for estimator  $\hat{\beta}_k$  can be written as:

$$t_k = \frac{z_k}{\sqrt{\frac{\hat{e}'\hat{e}/(N-K)}{\sigma^2}}} \quad \text{where we know:} \quad \frac{\hat{e}'\hat{e}}{\sigma^2} \sim \chi^2(N-K)$$

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  - ▶ What happens as  $N \rightarrow \infty$ ?
  - ▶ Mean  $\rightarrow 1$  and variance  $\rightarrow 0$
  - ▶ So,  $t_k \rightarrow z_k$  as  $N \rightarrow \infty$

## What about the $F$ distribution?

- Super simple—just the ratio of two scaled  $\chi^2$  distributions. Specifically:

$$F = \frac{X_1/k_1}{X_2/k_2} \quad \text{with} \quad X_j \sim \chi^2(j)$$

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- Suppose we have  $\beta$  is 3-dimensional, and we want to jointly test that  $\beta_1 = 0$  and  $\beta_3 = 0$ . That is, we want to test:

$$R\beta = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

## What about the $F$ distribution?

- Then, our test statistic is:

$$F = \frac{(R\hat{\beta})'[R(\mathbf{X}'\mathbf{X})^{-1}R']^{-1}R\hat{\beta}}{\hat{\sigma}^2} \quad \text{with} \quad \hat{\sigma}^2 = \frac{\hat{e}'\hat{e}}{N-K}$$

- Due to normality, under the null we know  $R\hat{\beta} \sim \mathcal{N}(0, \sigma^2 R(\mathbf{X}'\mathbf{X})^{-1}R')$
- So, divide top and bottom by  $\sigma$  and convince yourself we now have the ratio of two  $\chi^2$ s

## The best linear predictor

- Suppose our most basic model, with  $X$  a  $k \times 1$  random vector:

$$Y = X'\beta + U$$

- Recall that even with just our most basic regression assumption,  $E[XU] = 0$ , we know that the regression provides as the *best linear predictor* of  $Y$  given  $X$ . In particular, we can write:

$$BLP(Y|X) = X'\beta$$

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$$E[U * BLP(Y|X)] = E[UX'\beta] = E[UX']\beta = 0$$

This is of course true for every linear function of  $X$ .

- Key idea: BLPs are (by definition) linear functions of their conditioning sets.

## FWL: The BLP Interpretation

- Suppose now we partition  $X$  into  $X_1$  and  $X_2$ , where  $X_1$  is  $k_1 \times 1$  and  $X_2$  is  $k_2 \times 2$ , with  $k_1 + k_2 = k$ . Partition  $\beta$  accordingly. Then we have:

$$Y = X_1'\beta_1 + X_2'\beta_2 + U$$

- Suppose we want to recover  $\beta_1$  alone. Idea from FWL is:
  - ▶ First, residualize (project onto the null space of)  $Y$  and  $X_1$  on  $X_2$
  - ▶ Then, regress the residualized  $Y$  on the residualized  $X_1$ , which will recover  $\beta$
- Question: How do we actually execute this “residualization”?

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$$\tilde{Y} = Y - BLP(Y|X_2) \quad \text{and} \quad \tilde{X}_1 = X_1 - BLP(X_1|X_2)$$

- Then, consider the linear regression:

$$\tilde{Y} = \tilde{X}_1' \tilde{\beta}_1 + \tilde{U}$$

Where  $\tilde{U}$  is defined as per usual, as the residual in this new regression (so that  $E[\tilde{X}_1 \tilde{U}] = 0$ )

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- Task: Derive an expression for  $\tilde{\beta}_1$ .
- Hint: Remember that  $\tilde{Y}$  and  $\tilde{X}_1$  are residuals from a regression on  $X_2$

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- Although redlining is de-jure illegal today, there is still much debate about exactly *how* banks make decisions on whom to lend to, and whether those decisions create equitable outcomes
- Today, we'll look at a very basic version of an analysis from one of my papers, where we study how mortgage lending levels relate to income.
  - ▶ In future sessions, we may dive deeper into an analysis of federal policies designed to make lending more equitable
  - ▶ More info for those interested can be found here: <https://ssrn.com/abstract=4321265>

Now, we'll move to the sample R code to see how to actually apply FWL in regressions.

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- **True.** Note that:

$$\begin{aligned} E[Y|X] &= \text{Prob}\{X = 0\}E[Y|X = 0] + \text{Prob}\{X = 1\}E[Y|X = 1] \\ &= a(\text{Prob}\{X = 0\} + \text{Prob}\{X = 1\}) \\ &= a \end{aligned}$$

Then take the unconditional expectation of each side and apply the law of iterated expectations.

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We know that  $\beta = 0$ .

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- **False.** Without the constant, we have that:

$$\beta = \frac{E[XY]}{E[X^2]}$$

This is not the same as  $\frac{\text{Cov}[X, Y]}{\text{Var } X}$ , since while we know  $\text{Cov}[X, Y] = 0$  for independent r.v.s, we can't say the same for  $E[XY]$ . Counter example?

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$$\beta = \frac{E[XY]}{E[X^2]} = \frac{\text{Cov}[X, Y] - E[X]E[Y]}{\text{Var}[X] - E[X]^2} = \frac{0 - p^2}{p(1-p) - p^2} \neq 0$$

## Quick-fire problems #3

- Suppose  $X$  is a random variable with mean  $\mu$ . Suppose we are interested in the parameter  $\gamma = X_i \log(X_i)$ . For a sample of  $n$  i.i.d. draws of  $X$ , define the estimator:

$$\check{\gamma} = \frac{1}{n \log(n)} \sum_{i=1}^n X_i \log(X_i)$$

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- Note that:

$$\check{\gamma} = \frac{1}{\log(n)} * \frac{1}{n} \sum_{i=1}^n X_i \log(X_i)$$

We know  $\frac{1}{\log(n)} \rightarrow 0$  and, by the WLLN,  $\frac{1}{n} \sum_{i=1}^n X_i \log(X_i) \xrightarrow{P} \gamma$ . Therefore, by Slutsky,  $\check{\gamma} \rightarrow 0$ .

So  $\check{\gamma} \xrightarrow{P} \gamma$  iff  $\gamma = 0$ .